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MIDDLE VOLTAGE TRANSISTOR AND FABRICATING METHOD OF THE SAME

Abstract

A fabricating method of a middle voltage transistor includes providing a substrate. A gate predetermined region is defined on the substrate. Later, a mask layer is formed to cover only part of the gate predetermined region. Subsequently, a first ion implantation process is performed by taking the mask layer as a first mask to implant dopants into the substrate at two sides of the mask layer to form two first lightly doping regions. After removing the mask layer, a gate is formed to overlap an entirety of the gate predetermined region. Next, two second lightly doping regions are respectively formed within one of the two first lightly doping regions. After that, two source/drain doping regions are respectively formed within one of the two second lightly doping regions. Finally, two silicide layers are formed to respectively cover one of the two source/drain doping regions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a division of U.S. application Ser. No. 17/694,694, filed on Mar. 15, 2022. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a middle voltage transistor which having two lightly doping regions surrounding one source/drain doping region and method of fabricating the same.

2. Description of the Prior Art

[0003] Semiconductor devices are used in a variety of electronic applications such as personal computers, cell phones, digital cameras, and other electronic equipment. The fabricating method of a semiconductor device usually includes sequentially deposit materials of insulators, conductive layers, and semiconductor layers on a semiconductor substrate. Later, materials are patterned to by lithographic processes to form circuit elements on the semiconductor substrate.

[0004] As the integration of semiconductor elements increases, more elements can be integrated into a given area. However, as the feature size shrinks, additional problems arise in each process. For example, when the dielectric layer is removed, the gate dielectric layer is also etched. The etching of the gate dielectric layer leads to current leakage of the transistor completed afterwards.

SUMMARY OF THE INVENTION

[0005] In view of this, the present invention provides a new transistor structure and process so as to prevent current leakage.

[0006] According to a preferred embodiment of the present invention, a middle voltage transistor includes a substrate. A gate is disposed on the substrate. A gate dielectric layer is disposed between the substrate and the gate. A first lightly doping region is embedded in the substrate and extends to be under the gate. A second lightly doping region is embedded within the first lightly doping region, and the first lightly doping region surrounds the second lightly doping region, wherein the second lightly doping region includes a first edge. A source/drain doping region is embedded within the second lightly doping region, and the second lightly doping region surrounds the source/drain doping region, wherein the source/drain doping region includes a second edge. A silicide layer covers and contacts the source/drain doping region, wherein the silicide layer includes an end, and the end is disposed between the first edge and the second edge.

[0007] According to a preferred embodiment of the present invention, a fabricating method of a middle voltage transistor includes providing a substrate. A gate predetermined region is defined on the substrate. Later, a mask layer is formed to cover only part of the gate predetermined region. Subsequently, a first ion implantation process is performed by taking the mask layer as a first mask to implant dopants into the substrate at two sides of the mask layer to form two first lightly doping regions. After removing the mask layer, a gate is formed to overlap an entirety of the gate

predetermined region. Next, two second lightly doping regions are respectively formed within one of the two first lightly doping regions. After that, two source/drain doping regions are respectively formed within one of the two second lightly doping regions. Finally, two silicide layers are formed to respectively cover one of the two source/drain doping regions.

[0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 to FIG. 4 depict a fabricating method of a middle voltage transistor according to a preferred embodiment of the present invention, wherein:

[0010] FIG. 1 depicts a substrate having two lightly doping regions thereon;

[0011] FIG. 2 continues from FIG. 1;

[0012] FIG. 3 continues from FIG. 2; and

[0013] FIG. 4 continues from FIG. 3.

[0014] FIG. 5 depicts an enlarged top view of positions of a gate, first spacers, second spacers, an interface, a convex outer surface and an end of a silicide layer.

DETAILED DESCRIPTION

[0015] FIG. 1 to FIG. 4 depict a fabricating method of a middle voltage transistor according to a preferred embodiment of the present invention. As shown in FIG. 1, a substrate **10** is provided. A gate predetermined region A is defined on the substrate **10**. Later, a doping well **11** is formed within the substrate **10**. Subsequently, a mask layer **12** is formed to cover only part of the gate predetermined region A. The mask layer **12** may be photo resist, silicon nitride silicon oxynitride or other materials. In details, the gate predetermined region A has a symmetrical axis S. A symmetrical axis of the mask layer **12** overlaps the symmetrical axis S of the gate predetermined region A. That is, the mask layer **12** is in the middle of the gate predetermined region A. A width W of the mask layer **12** is smaller than the gate predetermined region A. Next, a first ion implantation process **11** is performed by taking the mask layer **12** as a mask to implant dopants into the substrate **10** at two sides of the mask layer **12** to form two first lightly doping regions LDD1.

[0016] As shown in FIG. 2, after the mask layer **12** is removed. A gate dielectric layer **14** is formed to blanketly cover the substrate **10**. Next, a gate **16** is formed on the gate dielectric layer **14**, and the gate **16** covers the entirety of the gate predetermined region A. Now, part of each of the first lightly doping regions LDD1 is directly under the gate **16**. In other words, part of each of the first lightly doping regions LDD1 overlaps the gate **16**. It is noteworthy that because the positions of the first light doped regions LDD1 are defined by the mask layer **12**, the overlapping region between the gate **16** and the first lightly doping regions LDD1 can be altered by adjusting the width W of the mask layer **12**. Next, two first spacers **18** are formed at two side of the gate **16**. Later, a second ion implantation process **12** is performed by taking the first spacers **18** and the gate **16** as a mask to implant dopants to penetrate the gate dielectric layer **14** to form the second lightly doping regions LDD2 in the substrate **10**. The second lightly doping regions LDD2 are disposed at two side of the gate **16**. One of the second lightly doping regions LDD2 is within one of the first lightly doping regions LDD1. The other one of the second lightly doping regions LDD2 is within the other one of the first lightly doping regions LDD1. The first lightly doping region LDD1 surrounds the second lightly doping region LDD2 disposed therein. Each of the first lightly doping regions LDD1 has a range greater than each of the second lightly doping regions LDD2.

[0017] As shown in FIG. 3, two second spacers **20** are respectively formed on one of the first spacers **18**. Next, a third ion implantation process **13** is performed by taking the gate **16**, the first

spacers **18** and the second spacers **20** as a mask to implant dopants to penetrate the gate dielectric layer **14** to form two source/drain doping regions S/D in the substrate **10**. The source/drain doping regions S/D are at two sides of the gate **16**. The source/drain doping regions S/D are respectively embedded within one of the second lightly doping regions LDD2. Moreover, the second lightly doping region LDD2 surrounds the source/drain doping region S/D embedded therein.

[0018] As shown in FIG. **4**, the gate dielectric layer **14** directly on the source/drain doping regions S/D is removed to expose the source/drain doping regions S/D. That is, the part of the gate dielectric layer **14** which is not covered by the gate **16**, the first spacers **18** and the second spacers **20** is removed. When the gate dielectric layer **14** directly on the source/drain doping regions S/D is removed, part of the gate dielectric layer **14** under the second spacers **20** is removed uncontrolledly. Therefore, a concave surface **22** is formed at an end of the gate dielectric layer **14** under the second spacers **20**. This concave surface **22** can even extend to be under the first spacers **18** sometimes. Furthermore, the concave surface **22** recessed in a direction toward the gate **16**. Later, a silicide process is performed to form two silicide layers **24** respectively cover and contact one of the source/drain doping regions S/D. The silicide layers **24** are not only formed directly on the source/drain doping regions S/D, but also fills into the concave surface **22**. That is, the silicide layers **24** will exceed the region directly on the source/drain doping regions S/D. Now, the middle voltage transistor **100** of the present invention is completed. The fabricating process of the middle voltage transistor **100** is suitable for manufacturing N-type transistors and P-type transistors.

[0019] FIG. **4** depicts a middle voltage transistor formed by the fabricating method provided in the present invention. The middle voltage transistor **100** of the present invention can be an N-type transistor or a P-type transistor. The middle voltage transistor **100** includes a substrate **10**. A first direction Y is perpendicular to a top surface of the substrate **10** and a second direction X is parallel to the top surface of the substrate **10**. A gate **16** is disposed on the substrate **10**. A gate dielectric layer **14** is disposed between the substrate **10** and the gate **16**. An end of the gate dielectric layer **14** includes a concave surface **22** recessed in a direction toward the gate **16**. A first lightly doping region LDD1 is embedded in the substrate **10** and extends to be under the gate **16**. A second lightly doping region LDD2 is embedded within the first lightly doping region LDD1, and the first lightly doping region LDD1 surrounds the second lightly doping region LDD2. The second lightly doping region LDD2 includes a first edge E1. A source/drain doping region S/D is embedded within the second lightly doping region LDD2, and the second lightly doping region LDD2 surrounds the source/drain doping region S/D, wherein the source/drain doping region S/D includes a second edge E2. The first edge E1 and the second edge E2 are both parallel to the first direction Y.

[0020] A silicide layer **24** covers and contacts the source/drain doping region S/D, wherein the silicide layer **24** includes an end **24a** and the end **24a** is disposed between the first edge E1 and the second edge E2. The end **24a** of the silicide layer **24** contacts the gate dielectric layer **14**. Moreover, two composite spacers **17** contact the gate **16** and the composite spacers **17** are respectively disposed at two sides of the gate **16**. Each of the composite spacers **17** includes a first spacer **18** and a second spacer **20**. The first spacer **18** contacts the gate **16** and is disposed at one side of the gate **16**. The other first spacer **16** also contacts the gate **16** and at the other side of the gate **16**. The second spacer **20** contacts one of the first spacers **16**. The other second spacer **20** contacts the other first spacer **16** which is at the other side of the gate **16**. Each of the second spacers **20** includes a convex outer surface **20a** that curves away from the gate **16**. Furthermore, an interface **19** is located at a contact region between the first spacer **18** and the second spacer **20** disposed at the same side of the gate **16**.

[0021] The first spacers **18** and the second spacers **20** are all entirely disposed at a top surface of the gate dielectric layer **14**, and a width of the gate dielectric layer **14** is greater than a width of the gate **16**. According to a preferred embodiment of the present invention, an entirety of the silicide layer **24** is not vertically under the first spacer **18**. It is noteworthy that the end **24a** of the silicide layer **24** is disposed between the convex outer surface **20a** and the interface **19**. FIG. **5** depicts an

enlarged top view of positions of a gate, first spacers, second spacers, an interface, a convex outer surface and an end of a silicide layer. As shown in FIG. 5, more specifically, as seen from a top view, the end **24a** of the silicide layer **24** is disposed between the convex outer surface **20a** and the interface **19**.

[0022] The width of the gate dielectric layer **14** and the width of the gate **16** are both parallel to the second direction X. The second direction X is not only parallel to the top surface of the substrate **10**, but also extends from one source/drain doping region S/D to the other source/drain doping region S/D. The width of the gate dielectric layer **14** equals to the summation of the width of the gate **16**, the width of the two first spacers **18** and the width of the two second spacers **20**.

[0023] Furthermore, the gate **16** includes a third edge E3 parallel to the first direction Y, the first lightly doping region LDD1 includes a fourth edge E4 parallel to the first direction Y. According to a preferred embodiment of the present invention, along the second direction X, a distance between the third edge E3 and the fourth edge E4 is smaller than half of the width of the gate **16**. Moreover, the operational voltage of the middle voltage transistor **100** can be operated at a normal operational voltage or an under drive voltage. For example, the operational voltage of the middle voltage transistor **100** can be between 1.8 and 9 volts. The thickness of the gate dielectric layer **14** is preferably between 100 and 250 angstroms.

[0024] The first lightly doping region LDD1 has a first dopant concentration, the second lightly doping region LDD2 has a second dopant concentration, the source/drain doping region S/D has a third dopant concentration. The third dopant concentration is greater than the second dopant concentration, and the second dopant concentration is greater than the first dopant concentration. For example, the first dopant concentration is between $1\text{E}17$ and $5\text{E}18\text{ cm.sup.-3}$, the second dopant concentration is between $5\text{E}18$ and $9\text{E}18\text{ cm.sup.-3}$, and the third dopant concentration is between $5\text{E}18$ and $9\text{E}20\text{ cm.sup.-3}$. The source/drain doping region S/D of the present invention is surrounded by two lightly doping regions (the first lightly doping region LDD1 and the second lightly doping region LDD2). The depth of the first lightly doping region LDD1 is greater than the depth of the second lightly doping region LDD2. The depth of the second lightly doping region LDD2 is greater than the depth of the source/drain doping region S/D. On the contrary, in a conventional transistor, the depth of the lightly doping region is smaller than the depth of the source/drain doping region S/D.

[0025] According to the fabricating process of the middle voltage transistor **100** mentioned above, the silicide layer **24** is not only formed directly on the source/drain doping regions S/D, but also fills into the concave surface **22** to make the end **24a** of the silicide layer **24** to extend to be under the second spacer **20**. If there is no second lightly doping region LDD2, the difference between the dopant concentration of the source/drain doping region S/D and the dopant concentration of the first lightly doping region LDD1 will be too large, and electrons will punch through at the concave surface **22**, and then current leakage occurs. Therefore, the second lightly doping region LDD2 which having a dopant concentration between the dopant concentration of the source/drain doping region S/D and the dopant concentration of the first lightly doping region LDD1 is specially provided in the present invention. Moreover, the end **24a** of the silicide layer **24** is in a position which does not exceed the first edge E1 of the second lightly doping region LDD2. In this way, current leakage can be effectively prevented.

[0026] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A fabricating method of a middle voltage transistor, comprising: providing a substrate, wherein a gate predetermined region is defined on the substrate; forming a mask layer to cover only part of the gate predetermined region; performing a first ion implantation process by taking the mask layer as a first mask to implant dopants into the substrate at two sides of the mask layer to form two first lightly doping regions; after removing the mask layer, forming a gate to overlap an entirety of the gate predetermined region; forming two second lightly doping regions respectively within one of the two first lightly doping regions; forming two source/drain doping regions respectively within one of the two second lightly doping regions; and forming two silicide layers to respectively cover one of the two source/drain doping regions.
 2. The fabricating method of a middle voltage transistor of claim 1, further comprising: before forming the gate, forming a gate dielectric layer to cover an entirety of the substrate, wherein the gate is disposed on the gate dielectric layer; after forming the gate, forming two first spacers respectively at two sides of the gate; performing a second ion implantation process by taking the two first spacers and the gate as a second mask to implant dopants to penetrate the gate dielectric layer to form the two second lightly doping regions; forming two second spacers respectively on one of the two first spacers; performing a third ion implantation process by taking the gate, the two first spacers and the two second spacers as a third mask to implant dopants to penetrate the gate dielectric layer to form the two source/drain doping regions; and before forming the silicide layer and after forming the two source/drain doping regions, removing the gate dielectric layer directly on the two source/drain doping regions.
 3. The fabricating method of a middle voltage transistor of claim 1, wherein each of the two second lightly doping region comprises a first edge, each of the two source/drain doping regions comprises a second edge, each of the two silicide layers comprises an end, and the end is disposed between the first edge and the second edge.
 4. The fabricating method of a middle voltage transistor of claim 3, wherein a first direction is perpendicular to a top surface of the substrate, and the first edge and the second edge are both parallel to the first direction.
 5. The fabricating method of a middle voltage transistor of claim 2, wherein a thickness of the gate dielectric layer is between 100 and 250 angstroms.
 6. The fabricating method of a middle voltage transistor of claim 1, wherein each of the two first lightly doping regions has a first dopant concentration, each of the two second lightly doping regions has a second dopant concentration, each of the two source/drain doping regions has a third dopant concentration, the third dopant concentration is greater than the second dopant concentration, and the second dopant concentration is greater than the first dopant concentration.
 7. The fabricating method of a middle voltage transistor of claim 6, wherein the first dopant concentration is between $1\text{E}17$ and $5\text{E}18 \text{ cm}^{-3}$, the second dopant concentration is between $5\text{E}18$ and $9\text{E}18 \text{ cm}^{-3}$, and the third dopant concentration is between $5\text{E}18$ and $9\text{E}20 \text{ cm}^{-3}$.
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